Mirza Nayeem Ahmed

github | linkedin | website | ahmed.mir@northeastern.edu | +8801552635263

Summary

Bridging data and social science to explore how technology shapes governance, policy, and inequality. Experienced in translating research into actionable insights to support evidence-based decision-making. Research interests include public opinion, behavior change, civic engagement, participatory HCI, and ICT for development.

Education

KU Leuven, Leuven, Belgium

Master of Science in Public Sector Innovation & eGovernance Sep 2026 - Present

A joint Erasmus Mundus Master's program co-funded by the European Union, delivered in partnership with the University of Münster (Germany) and Tallinn University of Technology (Estonia). The curriculum focuses on policy development, digital governance, and systemic technological change across the public sector.

Northeastern University, Boston, MA, USA

Bachelor of Science in Data Science Sep 2019 – May 2023

Minor: Entrepreneurship & Design Engineering

GPA: 3.66/4.0 Honors: Dean's List, Distinction

Relevant Courses: Natural Language Processing, Web Development, Object-Oriented Design, Discrete Structures, Network Science, Biostatistics, Database Design, Machine Learning, Human-Computer Interaction

Research

Microsoft Research **Jun 2024 – Jul 2025**

Pre-Doctoral Researcher | *Computational Social Science Lab*

Supervised by Dr. David Rothschild, Dr. Jake Hofman, Dr. Dan Goldstein

Modeling Public Political Perception Through Sociodemographic Lenses

- Contributed to a multi-paper research series on demographics, identity, and public opinion, supporting study design, manuscript preparation, and data pipeline development using Python.
- Applied clustering algorithms and exploratory models to large-scale survey datasets, uncovering significant patterns and emerging sociopolitical trends across population segments.

AI-Augmented Decision-Making in Online Prediction Markets

- Designed and applied predictive modeling and post-stratification methods to enhance demographic segmentation and improve the accuracy of 2024 U.S. Presidential Election forecasts.
- Benchmarked LLM performance for decision support in prediction markets (Polymarket, Kalshi), creating evaluation criteria to guide future research and tool development.

Longitudinal Study of AI Integration in Workplace Collaboration

- Leading a randomized controlled trial with sales and advertising teams to measure the long-term effects of AI-powered tools on productivity, KPI attainment, and workflow efficiency.
- Analyzing digital telemetry to track changes in collaboration patterns, tool adoption rates, and organizational knowledge-sharing behaviors over time.

University of South Florida **Nov 2023 – Jun 2024**
Research Associate | *Computational Social Science and AI Lab*
Supervised by Dr. Raiyan Abdul Baten

Modeling Creativity in Socio-Cognitive Systems
- Developed agent-based simulation models to analyze social, semantic, and temporal networks, exploring the influence of social interactions on belief rigidity and creativity within socio-cognitive systems.
- Engineered Python-based computational frameworks to simulate creative processes in social networks, integrating real-world data for enhanced accuracy and relevance.

Publications

- **M. N. Ahmed** and R. A. Baten, *From Semantic Memory to Collective Creativity: A Generative Cognitive Foundation for Social Creativity Models*, *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci 2026).*

Work Experience

Shastho-xyz (Dhaka, BD) Nov 2025 - Sep 2026
Data Consultant
- Conducting field research across hospitals, clinics, and diagnostic centers - interviewing clinicians, administrators, and patients - to map existing workflows and surface the institutional barriers shaping health data adoption
- Developing a long-term data strategy that examines infrastructure integration, lifecycle management, and financial feasibility, with attention to how legacy and paper-based systems define the constraints of digitization
- Studying how health data transitions play out across heterogeneous clinical environments, accounting for equity of access, regulatory constraints, and the gap between technical ambition and institutional reality

ANIQUE (Remote) Oct 2023 - Apr 2024
Data Consultant
- Led R&D initiatives to develop and implement an advanced, scalable data architecture for a dynamic e-commerce product catalog, ensuring secure and reliable cloud-based database operations
- Established strategic partnerships with external companies to acquire valuable data sources, facilitating data enrichment for informed business decisions
- Implemented AI algorithms for product attribution, search, and filtering to enhance UX and increase customer engagement

Condé Nast (New York, NY) Jun 2022 - Aug 2022
Data Analyst
- Built scalable ETL pipelines that support consumer events of the 16+ brands such as Vogue, GQ, WIRED, and The New Yorker for 2M+ customers using Apache Spark, Airflow, S3, and Databricks Delta Tables; enabling real-time data processing, providing visibility into critical metrics, and reducing data latency by 15%
- Applied scripting expertise to securely transfer sensitive information (database credentials, tokens, passwords, certificates, and API keys) to support vendor migration, enhancing data security and mitigating unauthorized access risks
- Leveraged Terraform to automate the provisioning and management of Condé Nast's global data platform, reducing infrastructure deployment time by 20% and providing a robust foundation for data processing and analytics
- Researched and developed a proof of concept for an in-house analytics tool to monitor social media engagement, securing buy-in from senior management (CTO, VP of Marketing, and SVP of Audience Development)

McKinsey & Company (Boston, MA) Jul 2021 - Dec 2021

Data Engineer

- Developed and implemented an automated recovery processing system in Domo to validate data, analyze usage, and calculate monthly chargeback for 15+ services, reducing manual steps and saving over 50% in ETL time
- Created and deployed a consolidated report visualizer for the finance and operation teams, centralizing critical data, providing real-time updates and ensuring data security and privacy compliance for over 200 employees
- Implemented AI-driven capabilities and leveraged external data sources to optimize a knowledge-aggregation tool for the Client Capabilities Hub, resulting in a 35% reduction in business analysts' information access time while ensuring data synchronization, CI/CD, and facilitating swift diagnosis and resolution

Teaching Experience

Microsoft Data Science Summer School 2025

Primary Instructor

- Led instruction in data science for students from underrepresented backgrounds exploring research pathways, with an emphasis on real-world data acquisition, cleaning, and analysis.
- Developed and taught modules on supervised and unsupervised learning, highlighting model evaluation, optimization, and practical applications such as recommendation systems and forecasting.

Northeastern University

Teaching Assistant

- Delivered over 400 hours of office hours, supporting 700+ students with assignments, projects, and technical concepts across diverse courses.
- Created grading rubrics and evaluated student quizzes and assignments at scale, ensuring consistency and transparency in assessments.

Skills

Data Science & Analytics	Python, R, Scala, Domo, Altair, Tableau
Experimentation & Surveys	Qualtrics, Prolific
Network Science	NetworkX, NetLogo
Project Management	JIRA, Confluence
Web Development	Figma, HTML, CSS, Java, Javascript, Typescript
Databases	MySQL, Oracle SQL, MongoDB, Redis, Neo4j
Data Engineering	Apache Spark, Airflow, dbt, Databricks, Snowflake, Alteryx

References

Dr. David Rothschild

Senior Principal Researcher, Computational Social Science Lab

Microsoft Research

davidmr@microsoft.com

Dr. John Rachlin

Associate Teaching Professor, Khoury College of Computer Sciences

Northeastern University

j.rachlin@northeastern.edu

Dr. Raiyan Abdul Baten

Assistant Professor, College of Engineering

University of South Florida

rbaten@usf.edu

Archan Bhatt

Director of Engineering & Associate Partner

McKinsey & Company

archan_bhatt@mckinsey.com